

Vibration Signal-Based Fault Prediction Using Few-Shot Learning in the Digital Twin Ecosystem

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Keywords: Digital Twin, Condition Monitoring, Predictive Maintenance, Intelligent Manufacturing, Smart Maintenance

Abstract

The current era is the era of digital transformation. Various concepts such as Cyber-Physical Systems, digital twins, etc. are proposed to implement this transformation. A digital twin is a virtual replica that can monitor and predict the state of any asset or process enabling the use of various cutting-edge technologies. A digital twin has a variety of applications. Maintenance is one area where digital twins are actively used for predictive maintenance. To work efficiently a real time data collection mechanism is required. Also, it may not always be feasible to capture and train large data sets for Machine learning models due to the cost and other limitations associated with it. In this paper, a digital twin is created using OPCUA server. Digital Twin captures the data from the IoT sensors with the help of various protocols. This data is intelligently analyzed with various machine learning algorithms built into the digital twin to monitor health status. Few-shot learning, an algorithm in which a machine learning model can learn with a limited amount of data and perform reasonably accurately is used. For this study, a rotary asset is explored for fault prediction. The vibration signals are converted into 2D images which are then trained on CNN models using the Siamese Neural Networks method. The performed analysis proven to be very effective in various fault classifications. The knowledge gained through few-shot learning remains with digital twins to monitor similar kinds of newly added assets without any much training.

1 Introduction

Industry 4.0 represents the fourth industrial revolution which is supported by the integration of AI/ML into manufacturing and production processes [1]. A key concept within Industry 4.0 is the "digital twin," a virtual representation of a physical product, process, or system [2]. Digital twins leverage real-time data of the physical asset or processes to simulate, monitor, and optimize performance [3], enabling predictive maintenance. The digital twin collects data from various sensors or controllers using several machine-to-machine communication protocols such as OPCUA [4] and MT Connect [5]. In the context of maintenance, digital twins use several machine learning and deep learning algorithms to provide insights into equipment health, early issue detection, and efficient scheduling of maintenance activities [4]. This technique can enhance operational efficiency, reduce downtime, and transform traditional industry maintenance practices. These prognostic health predictions are inferred from analyzing various physical characteristics such as vibration signals, acoustic signals [6], and current waveforms [7]. The selection of suitable attributes and AI/ML algorithms depends on the type of data and its accessibility. These algorithms use data-driven insights to generate predictions about the health status of machinery, helping in reducing downtime.

Several methodologies have been developed for the analysis of vibration data. These techniques are well-known and studied at length in literature [8]. These techniques include the analysis of vibration data via various statistical parameters. These statistical parameters are calculated for the time

domain, frequency domain, and time-frequency domain. Extensive research is being done to extract these features, as they have valuable insights into the condition of assets. Various algorithms are available to calculate various statistical features. These features are fed into the AI/ML model for model training [8]. However, the extraction of insights from data sets riddled with noise presents a challenge. Another notable and innovative method involves the transformation of conventional 1D vibration signal data into a 2D image format [9]. This can be achieved either by the thermographic image [10], a noninvasive technique or by converting vibration signals into 2D image signals. The vibration sensor signal to 2D images is under the scope of the study of this paper. This conversion can be done by utilizing various techniques such as FFT spectrum image [11], Short Time Fourier Transform (STFT) spectrograms [12], and the RGB vibration image representation (RGBVI) [13], etc. to convert into the bi-dimensional image. These 2D images are analyzed using various ML/DL techniques such as Computer vision, Convolutional Neural Networks (CNN), or any other complex deep learning algorithms. An exhaustive list of various techniques studied in literature is presented in [14] to convert vibration sensor signals into 2D images and further analysis. However, the data requirements are still very high in these methods. It may not always be possible and feasible to collect large amounts of data. Thus it may not be suitable to apply all these techniques everywhere.

In this paper, a technique is used to minimize the requirement of data. The learning of a pre-trained open image model via transfer learning is used. This has further pertained via using the Siamese network. Finally, the pre-trained model is used

with the SVM algorithm to accurately classify the type of fault present. By doing so, data requirements are minimized. A rotary equipment is studied by inducing four different types of faults in it. Vibration signals are converted into 2D images for further analysis. A total of 16 types of faults have been induced by applying a full factorial experiment.

The remaining paper is divided into the following sections. The first section describes the use of techniques for data acquisition. Within the scope of this study, an OPCUA server has been developed to collect real-time, high-frequency data streams. Subsequently, the second section explores the conversion and 1D image to 2D image data. The third section explains the training and testing of SVM algorithms using weights obtained in the pre-training. Finally, the last section offers a detailed analysis of the results and is followed by a conclusion.

2 Data Collection Using IoT Sensor via OPCUA

A health monitoring system in a digital twin ecosystem comprises three primary phases: Data collection, data pre-processing, and health diagnosis. In the scope of this study, the vibration data is collected through the use of an IoT device. This IoT device is designed and developed by the authors within their laboratory environment. The development of the device for vibration data includes the integration of a vibration sensor with an ESP32 microcontroller. An integral part of this setup is the creation of an OPC UA server, which is embedded within the microcontroller. This OPCUA server operates with high efficacy to enable the collection of high-frequency vibration signal data in real-time over the network. OPCUA server is a socket-based, bidirectional Machine-to-Machine industrial communication protocol allowing IoT devices to share data at very high frequency. However, this server also requires an OPCUA client. This server and client communicate using a server-client mechanism. The client collects data wired/wirelessly at a high frequency and is made available in the cloud for further detailed analysis.

Once the IoT device setup was ready, the data collection process was started. A total of four types of faults were introduced into the rotary asset. These faults were: mass imbalance, motor bolt looseness, fan bolt looseness, and fan

blade cage looseness. The vibration signals were recorded at 2000 Hz frequency.

Table 1 A 4-bit code is used to represent the Full Factorial experiment of data collection during the experiment

4-bit code	Mass imbalance (MI)	Bottom bolt looseness (BBL)	Motor bolt looseness (MBL)	Front cage looseness (FCL)
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
...	...	...	...	...
15	1	1	1	1

The data acquisition was performed by taking two levels for each fault as shown in Table 1. Level 0 indicates the corresponding fault was not present and Level 1 indicates that the fault was present. All combinations of faults, leading to a full factorial experiment, were performed resulting in 16 different fault classes. These 16 fault classes were indicated by a 4-bit number and are summarized in Table 1.

3 Data Processing

This section discusses the pre-processing techniques of the Mel spectrogram used in our analysis. This is followed by feature extraction using Convolutional Neural Networks (CNN), and then using trained weights for training on SVM classifiers. Mel spectrogram serves as an important tool in our pre-processing phase, translating vibration signals into a binary image. This binary representation is then further transformed into an RGB image, providing a more comprehensive visual representation of the underlying data. As shown in Figure 1 healthy vibration signals are represented in the RGB image. Figure 1, shows a set of images, each representing the acceleration experienced in the x, y, and z directions. For each direction, three different samples are displayed. Thus, it becomes evident that the vibration pattern in each direction shows a characteristic uniqueness, and this uniqueness is consistently observed across all three samples.

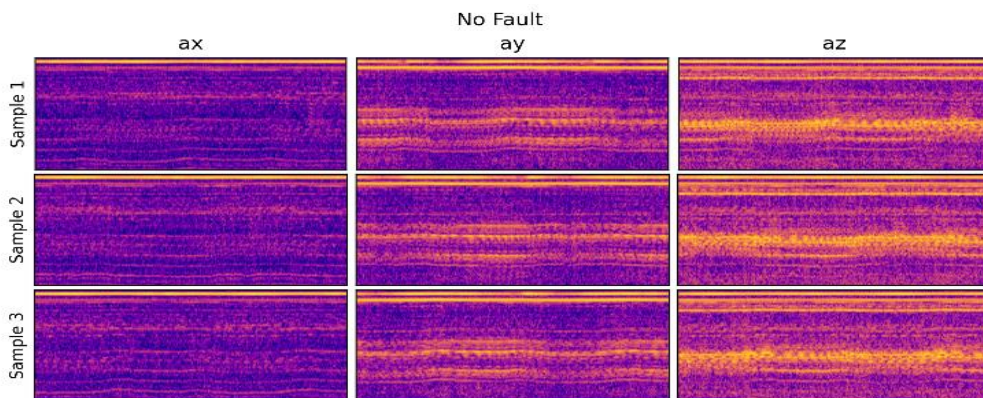


Figure 1 Vibration signal to RGB Image (with no fault)

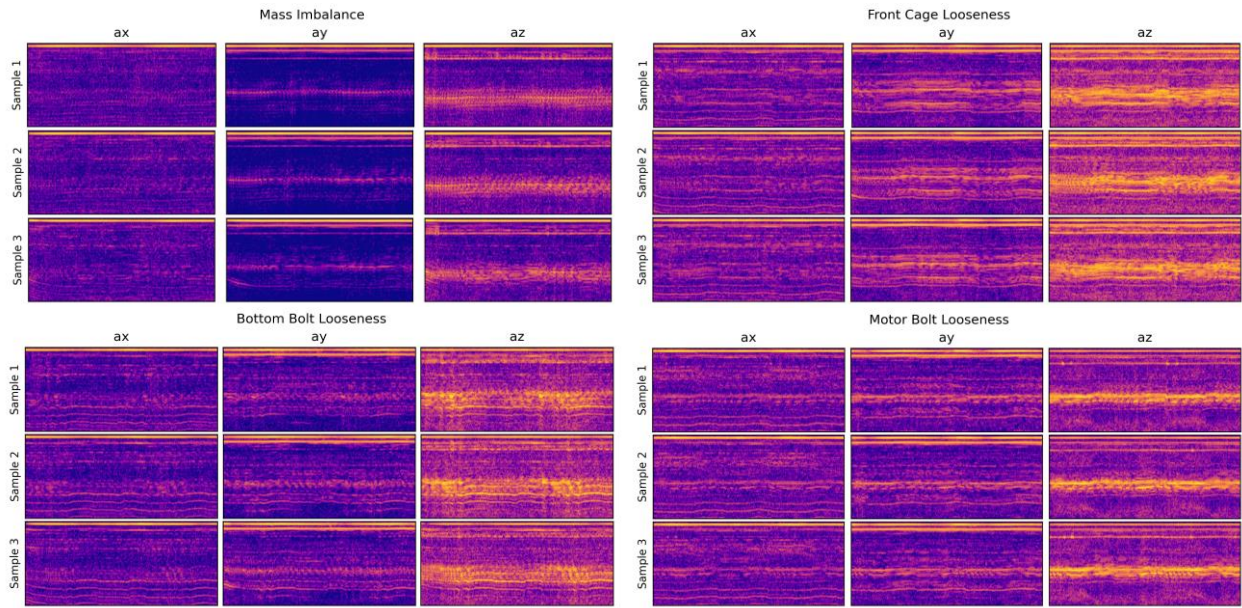


Figure 2 Different Vibration signal image correspondence to various fault

Figure 2, presents a visual representation of four distinct fault types: mass imbalance, front cage looseness, bottom bolt looseness, and motor bolt looseness. As it is clear, each of these fault types shows a distinct pattern within the provided images. This unique pattern is the key identifier for the specific fault type identification.

A deep learning model can be trained for feature learning and further classification based on these unique patterns of each fault. To develop the architecture of deep learning models, a

number of architectures and algorithms are reviewed.

4. Training and Testing of Deep Learning Model

The detailed architecture of the deep learning model is shown in Figure 3.

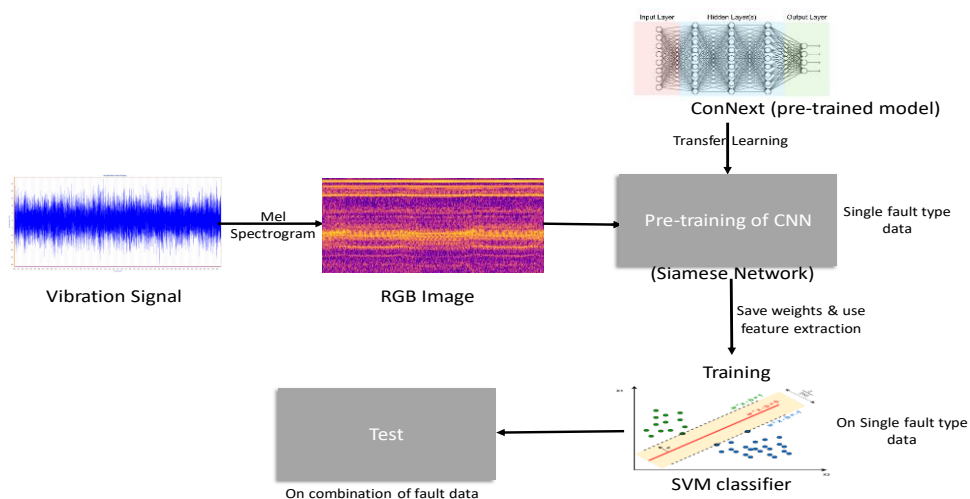


Figure 3 Detailed Architecture of the model

According to this architecture, vibration signal data are converted into the RGB image using the Mel spectrogram. The learning of a pre-trained open-source model ConNext is used via transfer learning.

A CNN model is used to utilize the learning of the ConNext

model to pre-train our model on experimental data sets. The vibration signals in the form of RGB images are used to pre-train the model, using the Siamese network, to learn the important features as shown in Figure 4. Siamese Network algorithms find application in situations with extremely

limited training data. They are particularly well-suited for scenarios where deploying sensors or gathering data is resource-intensive.

In the training process of a Siamese network, two images are compared as shown in Figure 4 and their dissimilarity is quantified in the form of distance matrices. Using this distance information, it is determined whether both images belong to the same class or not. Based on the loss function, weights are updated. This is performed on a total of 12 samples out of which 9 are used for training and the remaining 3 are used for validation. The pre-training is performed on single-fault data (no fault combination). The trained weights are stored for subsequent utilization in our feature extraction process. These weights are obtained through the Siamese network training. Once it is done, we start training the model to identify actual machine faults. This is achieved through the SVM classification algorithm. The SVM model is also uses the same 12 samples out of which 9 for training and 3 for testing. The SVM model is also trained on a single fault (without any combination of fault in data) in data sets. The trained SVM network has proficiency in accurately identifying and categorizing various machine faults. It is important to note that the trained SVM network showcases impressive results by achieving the capability to predict faults with various combinations. This proves the effectiveness of our designed model for real-world scenarios.

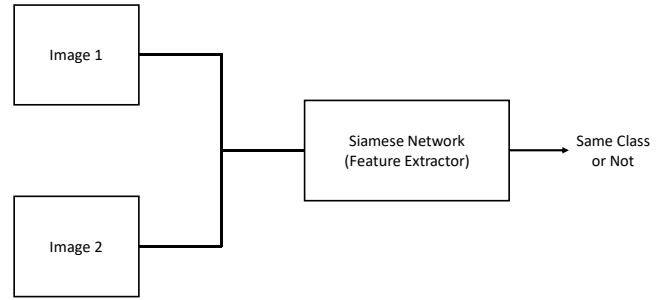


Figure 4 Pre-training on the data set using Siamese Network

5. Results & Discussion

The results and discussion section demonstrate the effectiveness of the developed Deep Learning model for fault classification. The model was trained using sample datasets, utilizing the SVM algorithm, as described in the model architecture. Figure 5 explains that the model accurately classifies different fault classes during the training process. During training, the weights obtained were subsequently used for testing. Notably, the training process was conducted exclusively on single fault-type data. However, in the testing phase, the model is able to classify both single faults and combinations of faults, as shown in Figure 6. This proves the model's robustness and its capability to identify faults even in scenarios where multiple faults may be present simultaneously.

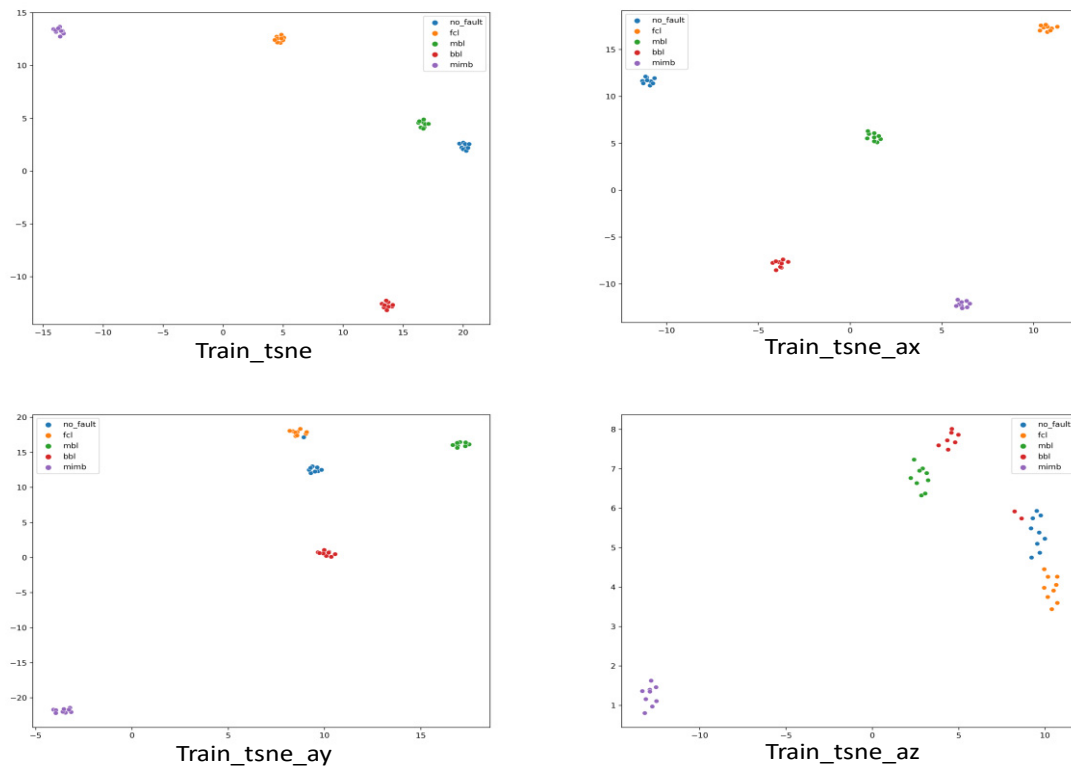


Figure 5 Training on single fault data

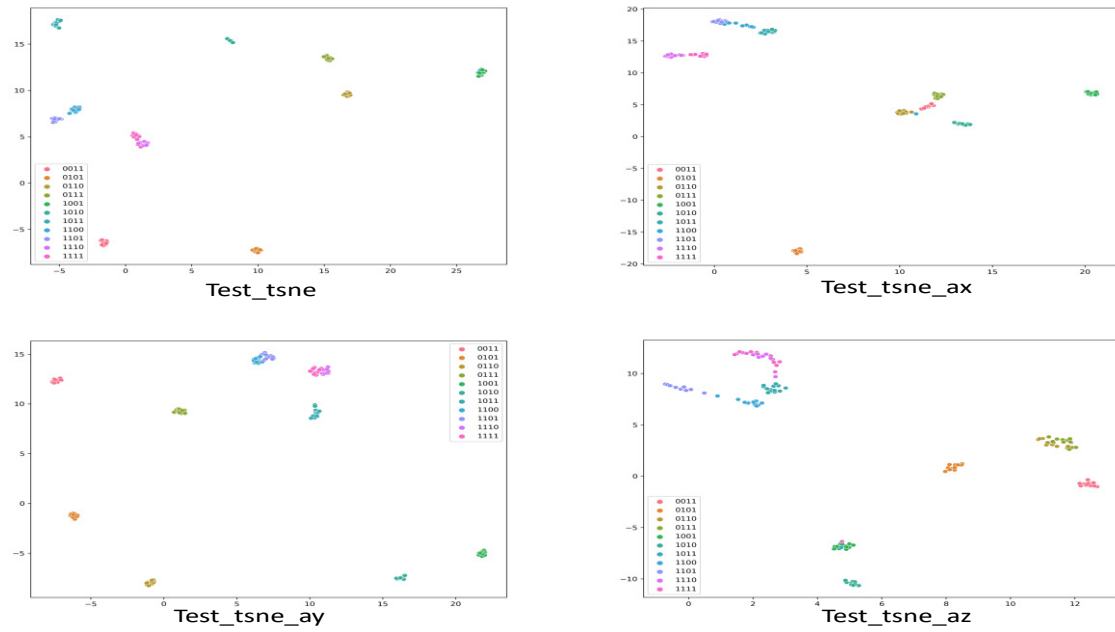


Figure 6 Testing on single and multiple fault data

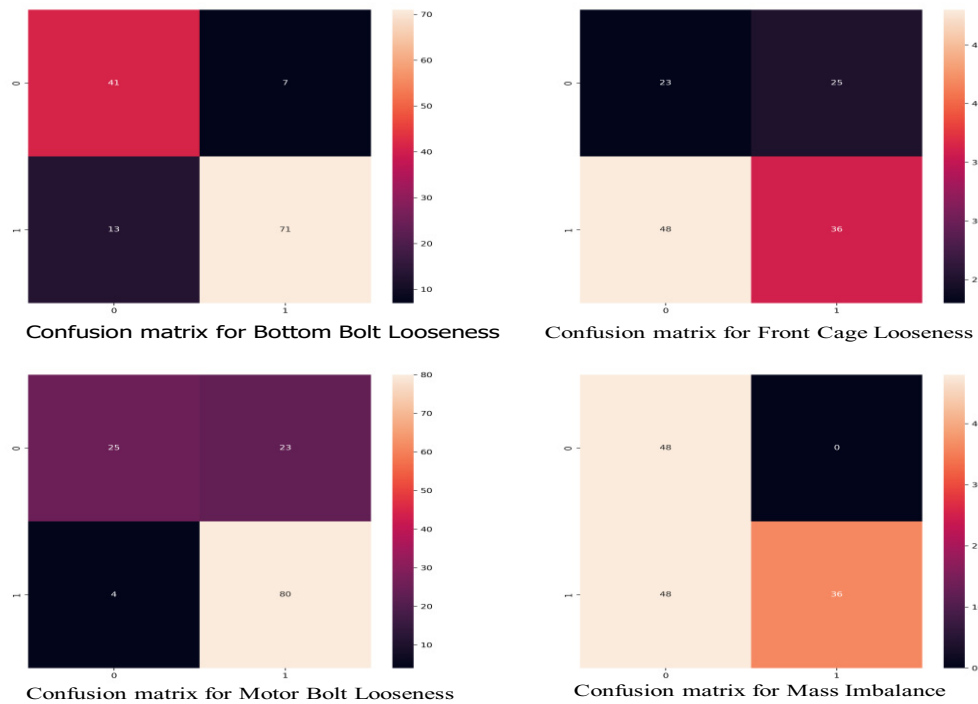


Figure 7 Confusion matrix for all fault types

Figure 7 shows the confusion matrix, which serves as a visual representation of the classification model's performance. For instance, in the case of bottom bolt looseness, the model made 112 correct predictions, whereas for front cage looseness, it achieved 59 correct predictions. Similarly, for motor bolt looseness, the model correctly predicted 105 instances, and for Mass imbalance, it achieved 84 correct predictions. These prediction results highlight that the model excels in detecting

cases of bottom bolt looseness and motor bolt looseness. This outcome proves the effectiveness of our model, as these specific fault types were particularly distinguishable in the vibration signature data used for training and testing. The overall findings affirm the model's capability to accurately classify various fault types, proving its potential for real-world application in fault detection and diagnosis scenarios.

6. Conclusion

A digital ecosystem is where all the assets are digitally connected. These connected digital assets share the data over the cloud in real-time. These assets are continuously monitored for various diagnosis activities. A condition-based monitoring system has been created to monitor the health of a rotary asset within a digital twin ecosystem. To achieve the desired aim of the system, data acquisition is a prerequisite. To achieve this, an OPCUA server was created to facilitate data collection. Data was collected under different fault conditions, enabling the development of a deep-learning model for fault prediction. Data collection may not be feasible in all the scenarios; thus the architecture of the model is designed to minimize the data requirements. The results prove the efficacy of the designed model on the test datasets. Simultaneously, the digital twin can now leverage this model for monitoring similar types of assets under it. In the future scope, a framework for the digital twin ecosystem can be designed where the learning achieved in this study can be applied to other similar/dissimilar kinds of assets.

7. References

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